

USER GUIDE

firmware 2.3 for the M-VAVE FM-1

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Welcome to the dungeon

GHOULBOX turns the M-VAVE FM-1 into a dungeon-synth instrument: hurdy-gurdies and drones, crypt pads and monk choirs, organs, harps and recorders, all through a cathedral reverb and a worn cassette.

It is a fork of **Felucca**, the open-source FM-1 firmware by Hügelton Instruments. Everything Felucca does still works here: four tracks, a 64-step sequencer, songs, the arpeggiator, chord keys, effects, the web editor and backups. GHOULBOX adds the sounds, the reverb and tape, a hurdy-gurdy engine, dungeon drums and an old-RPG look, and leaves out the sounds that don't fit the genre.

How to use this guide

- **New to synthesizers?** Read Part I to get the FM-1 running, then work through Part II. Each chapter there ends in a **Try this** box: a few steps on the instrument that make the idea audible. Nothing you do in them can break anything, and holding `SAVE` undoes the last sound or pattern load.
- **Know synths already?** Part III is the reference: every page, every sound, every setting.
- **Want to make a piece?** Part IV walks through one from an empty track to a recording, with a chord cheat sheet.
- **Rather watch?** Five short video lessons, on the FM-1's own screen, go from a single drone to a finished piece: `youtube.com/playlist?list=PLMUs0QD7QfOw` (or search YouTube for GHOULBOX Lessons).

Buttons and knobs are written like `PLAY`. A path such as `SEQ` › STEP means "press SEQ until the STEP page shows".

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> GHOULBOX is a hobby project, tested on one FM-1. Install it at your own risk. Save a backup in the web editor first (the installer doesn't), and you can always go back to Felucca or M-VAVE's own firmware.

PART I

The instrument

What the FM-1 is, how to put GHOULBOX on it, and how to find your way around its knobs, buttons and screen.

1. Meet the FM-1
2. Install, back up, go back
3. Five-minute start
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1 · Meet the FM-1

The FM-1 is a small battery-powered synthesizer: a keyboard, a screen, eight knobs, a speaker and a computer chip that makes the sound. GHOULBOX is the program that runs on that chip.

M-VAVE sells the FM-1 with its own firmware, an FM synthesizer. GHOULBOX replaces that program completely, so most of M-VAVE's manual (its pages, presets and buttons) does not apply. The hardware is the same, though, and this chapter covers it.

On the top

- **27 keys** (silicone, about two and a quarter octaves), each with a light under it. They are not velocity-sensitive: every key plays at the same strength. A MIDI keyboard can add velocity (chapter 20).
- **MASTER**, the volume knob. It has a start and an end, like a volume knob on a radio.
- **Seven endless knobs**: SELECT, PRESETS, ALGORITHM and KNOB 1–4. They turn forever in either direction and move a value in clicks.
- **Buttons** for pages, transport (PLAY, REC) and octave (OCT−, OCT+).
- **A 1.54-inch colour screen**.
- **A speaker**, small but enough to practise with.

On the back

Socket	What it's for
Audio out (3.5 mm)	Headphones, or a cable to speakers, a mixer or an audio interface. Plugging something in silences the speaker. Use a normal stereo (TRS, two black rings) plug: a phone-headset plug with three rings can buzz and may not silence the speaker.
MIDI IN (3.5 mm)	A cable from another instrument's MIDI OUT, through a small 3.5 mm-to-MIDI adapter. M-VAVE's manual doesn't say which of the two adapter wirings (called Type A and Type B) it uses; if a keyboard plays nothing through it, try the other type.
USB-C	Charging, and the connection to a computer: MIDI both ways, the FM-1's sound as a USB audio input, and the installer and editor.
Power switch	On and off.

Power and battery

- A built-in 2000 mAh battery. M-VAVE quotes about 12 hours of play; that figure is for its own firmware, and GHOULBOX's reverb works the chip harder.
- Charge it from any USB-C charger or computer port. It also runs from USB power.
- The battery level is in the top-right corner of the screen.

No Bluetooth

M-VAVE's firmware offers wireless (Bluetooth) MIDI. Felucca and GHOULBOX don't: use USB or the MIDI IN socket instead.

Specifications

Size	161.5 × 96.5 × 28.6 mm
Weight	251 g
Battery	3.7 V lithium-polymer, 2000 mAh, charged over USB-C
Sound under GHOULBOX	44.1 kHz stereo; four tracks sharing 12 voices

> Look after it as you would a phone: keep it dry, don't leave it in a hot car, and charge it now and then if it sits unused for months, so the battery never runs completely flat.

2 · Install, back up, go back

> Try it first: `ghoulbox.messy.quest` plays GHOULBOX in your browser, the same firmware on a drawn FM-1, with your computer keyboard as the keys.

What you need

- A computer with **Chrome or Edge** (other browsers can't talk MIDI to the FM-1)
- A USB **data** cable (some cables only charge)
- No other music software open (it can hold on to the FM-1's MIDI port)

Install

1. Connect the FM-1 and switch it on. **Save a backup first**, because the installer doesn't make one: open the web editor at `ghoulbox.messy.quest/webapp/editor`, allow MIDI access, and on the **Settings** tab press **Save backup**. Your music, sounds, samples and settings go to your computer.
2. Open `ghoulbox.messy.quest/webapp/installer` and press **Install**. Allow MIDI access when the browser asks.
3. The installer switches the FM-1 to update mode and writes GHOULBOX.
4. When it says done, the FM-1 restarts with the GHOULBOX skull. Hold `HOME` › SYSTEM › ABOUT shows the version.

> If the installer stops after it says "update mode", unplug the USB cable, plug it back in, and press Install again. The FM-1 stays in update mode until the install is finished, so nothing is lost.

Updating later

Save a backup in the web editor, then run the installer again. Your projects and sounds stay where they are.

Going back

To return to Felucca, install it from `hugelton.github.io/Felucca` and restore your backup in its editor. Its **Return to official V15** button puts back M-VAVE's own firmware.

Your old music

Projects and user presets from Felucca load in GHOULBOX. Sounds GHOULBOX no longer lists (for example the techno presets) still load and play inside an old project; they are only missing from the menus.

3 · Five-minute start

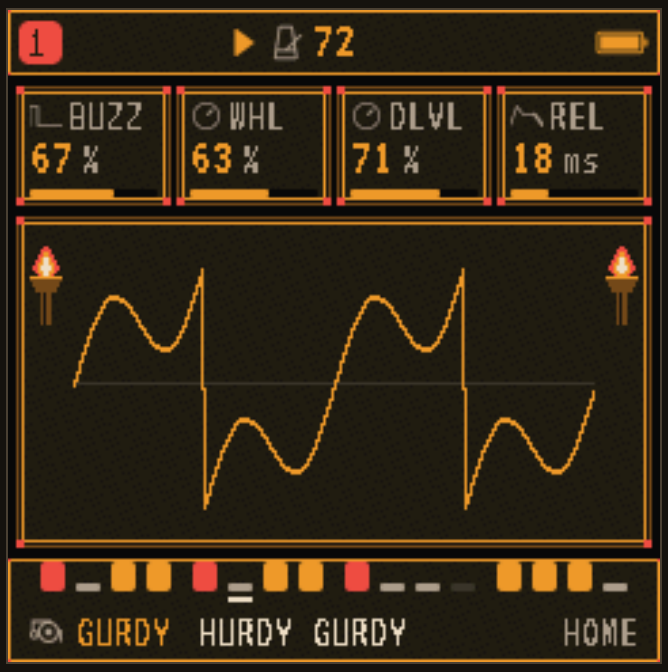
GHOULBOX powers on ready to play a small dungeon piece.

1. **Press `PLAY`.** Four tracks play a scene in D minor at 72 BPM: a hurdy-gurdy melody (track 1), a crypt pad (2), chanting monks (3) and a drone (4), through the hall reverb. Press PLAY again to stop.
2. **Turn `MASTER`** for volume. Headphones in the jack silence the speaker.
3. **Turn `ALGORITHM`** to choose the track (T1–T4). Everything you do applies to that track.
4. **Turn `PRESETS`** to change the track's sound while it plays. Your pattern stays.
5. **Play the keys** to play the selected track. `OCT-` / `OCT+` shift the octave.
6. **Turn `SELECT`** to change the tempo.
7. **Turn `KNOB 1-4`** to shape the sound: on HOME they are the engine's four main controls.

> Made a mess? Hold `SAVE` to undo the last sound or pattern load. Switching the FM-1 off and on brings back what you were playing (stop and leave it for a few seconds first, so it can save it).

Other scenes

This is the **DUNGEON** scene. Hold `HOME` for the menu, and on SYSTEM › **SCENE** choose the scene a new FM-1 starts with (or every power-on, with SYSTEM › RESTORE LAST set to OFF): **CRYPT** (a low drone, a crypt choir, cave wind and slow tomb drums, 60 BPM), **TAVERN** (a dance gurdy, a harp, a lute and a light drum kit, 96 BPM), **CHAPEL** (cathedral organ, monks, a recorder and a fiddle, 66 BPM), or **BLANK** (DUNGEON's sounds with no notes, for starting from nothing).



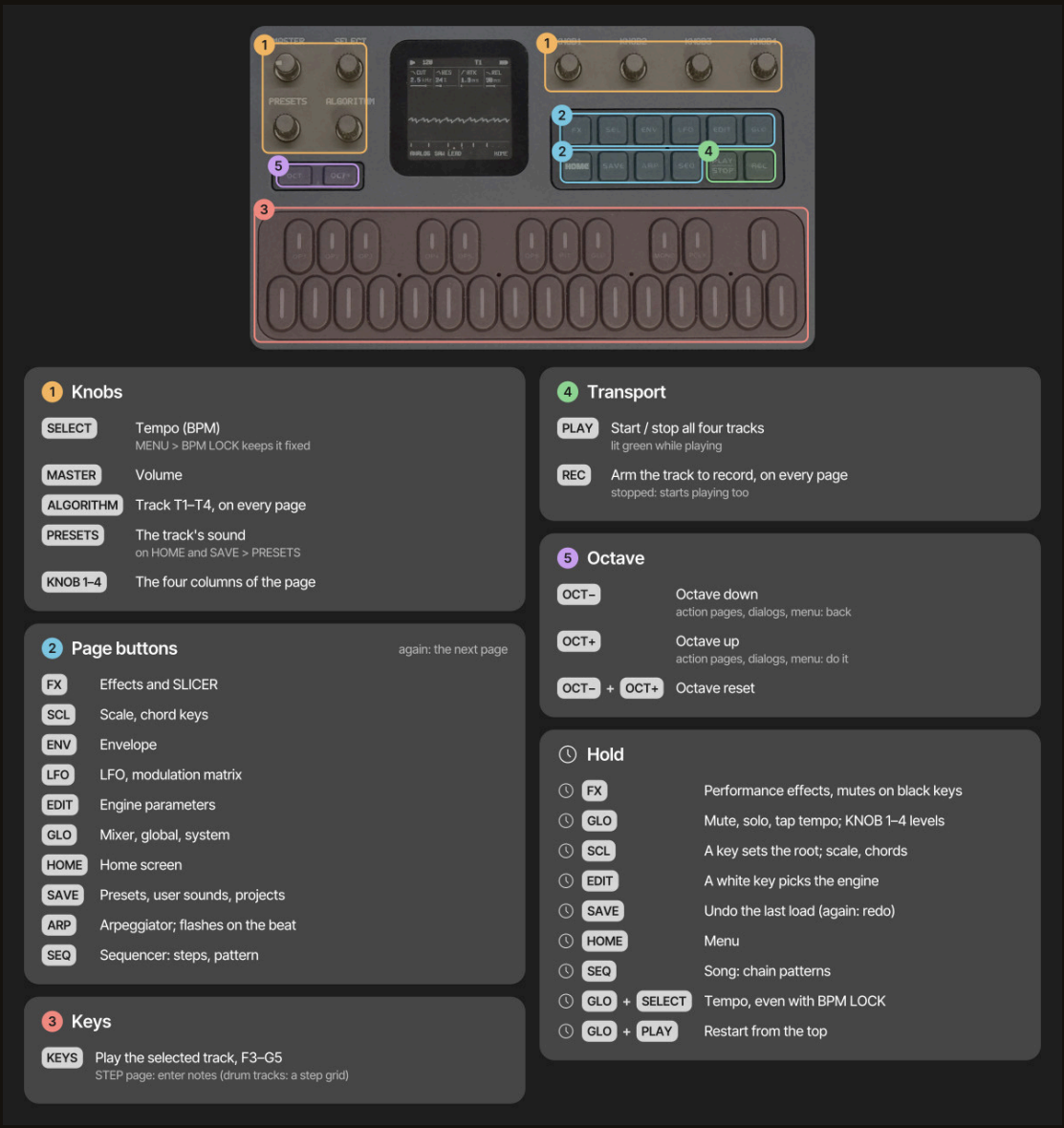
HOME: four knob cards, the scope between the torches, the track's steps and sound at the bottom



SEQ › STEP: the piano roll of the selected track

4 · The panel

The FM-1's controls, as Felucca (and GHOULBOX) use them. The labels printed on the panel are M-VAVE's; most still fit, and the diagram shows what each one does now.



Panel diagram from Felucca's documentation (Hügelton Instruments, GPL-3.0). The screen in it shows Felucca's look; GHOULBOX's is in the next chapter.

Three ways to press a button

- **Tap** a page button for its page, and again for the next page in its group (chapter 14 lists them all). `HOME` returns to the home screen.
- **Hold** `FX`, `GLO`, `SCL`, `EDIT`, `SEQ` or `REC` for a **quick layer**: shortcuts on the keys and knobs, with a map on the screen. Let go to close it.
- **Double-tap** one of them to keep its layer open without holding; tap it again to close.

Other holds: `HOME` opens the menu and `SAVE` undoes. What the layers hold:

- **FX**: performance effects on the keys (chapter 16).
- **GLO**: the black keys F♯ G♯ A♯ C♯ of the lower octave mute tracks 1–4, the white keys F–B solo them while held, C unmutes all, the upper F taps the tempo, and the upper F♯ G♯ A♯ C♯ latch a drone (chapter 15). KNOB 1–4 are the four tracks' levels; SELECT is the tempo even when BPM LOCK is on.
- **SCL**: press any key to make it the ROOT; KNOB 1–4 are ROOT, SCL, CHRD and VOIC.
- **EDIT**: each white key picks an engine.
- **SEQ** (SEQ TOOLS): the white keys from the lowest F are tools for the selected track's steps: CLEAR, REVERSE, SHIFT ‹, SHIFT ›, RANDOM and COOK (a few notes changed). KNOB 1–4 are LEN, DIV, SWG and GATE (chapter 13).
- **REC**: the white keys from the lowest F are CLEAR (the selected track's steps and automation, at once; hold SAVE to undo), CLICK, COUNT-IN and CLICK LEVEL; each press steps to the next value.

Yes and no

Wherever the FM-1 asks something (a dialog, an action page), `OCT+` means yes / do it and `OCT-` means no / go back. Their lights show when they answer.

5 · The screen

GHOULBOX draws Felucca's pages as an old role-playing game would: stone windows, a pixel font, flickering wall torches beside the scope, and a **sword** that points at what is selected.



PRESETS: the sword marks the current sound



The MENU (hold HOME) in five tabs

- **The header** shows play state, tempo, the track number and battery.
- **Knob cards**: most pages show four cards in a row, one for each of `KNOB 1-4`, left to right. Each has a name, a value and a small gauge. The one you just turned is drawn in the accent colour.
- **The scope** on HOME draws the sound itself as a moving line: smooth for soft sounds, jagged for bright ones.
- **Messages** (for example SOUND INIT, STOP TO SAVE or NO SAMPLE) appear in the header for a moment.
- **Colours:** a new FM-1 starts in **CRYPT** (stone, bone, torch amber, blood red). If you came from Felucca, your saved palette stays: hold `HOME` and pick COLOR › CRYPT.

The lights

Idle buttons and keys glow dimly so you can find them in the dark. A button that does something right now breathes slowly. `PLAY` turns green while playing, and the keys light up with the notes being played, whether by you, the sequencer or MIDI. MENU › LEDS sets how bright the idle glow is.

PART II

Synth basics

How a synthesizer makes and shapes sound, in plain words. Each chapter ends in a short exercise to try on the FM-1. Read them in order: each builds on the one before.

6. What a synthesizer does
7. Notes, keys and scales
8. Tone: oscillators and the filter
9. Shape in time: the envelope
10. Movement: the LFO and modulation
11. Voices: poly, mono and glide
12. Space: effects
13. Time: tempo, steps and the arpeggiator

6 · What a synthesizer does

Every sound is air shaking back and forth. A synthesizer is a machine that decides exactly how it shakes.

Any sound you hear differs from any other in three ways:

- **Pitch**: how high or low. A fast shake is a high note, a slow one a low note.
- **Loudness**, and how it changes: a harp note starts loud and fades; an organ note stays the same while held.
- **Tone** (or *timbre*): what makes a flute and a trumpet sound different on the same note. Bright sounds carry many high overtones; dark, soft sounds carry few.

A synthesizer has a part for each of these. Most of them, including the FM-1, are built the same way:

```
OSCILLATOR → FILTER → AMPLIFIER → EFFECTS
```

- The **oscillator** makes the raw sound at the pitch of the key.
- The **filter** shapes the tone, usually by removing brightness.
- The **amplifier** sets the loudness over time, shaped by an **envelope**.
- **Effects** place the sound in a space: reverb, delay, chorus, tape.

On top of that sit **modulators**, which move the other parts by themselves (a slow wobble, a swell), and a **sequencer**, which plays notes for you.

Engines

The FM-1 under GHOULBOX has **13 engines**: thirteen different ways of making the raw sound. ANALOG copies the classic synthesizers of the 1970s and 80s; VOICE sings vowels; GURDY models a hurdy-gurdy; SAMPLE plays recordings of real instruments. Each engine comes with ready-made settings called **presets** (or sounds), 59 of them in all.

Tracks

GHOULBOX has **four tracks**, like four musicians. Each has its own engine and sound, its own notes in the sequencer, and its own settings. The `ALGORITHM` knob chooses which track you are playing and editing; the screen shows T1–T4. All four share **12 voices** (12 notes sounding at once), so a big chord on one track can cut short a long note on another.

TRY THIS · Hear the parts

1. Press `PLAY`. Turn `ALGORITHM` to T2 (the pad).
2. Hold `GLO` to open its layer. While holding it, the first four black keys (from F♯ in the lower octave) mute and unmute tracks 1–4; their lights show which are sounding. Mute all but track 2 and listen to the pad alone, then bring the others back one at a time.
3. With T2 still selected, turn `PRESETS` slowly: the pad's part plays on a different sound each time, while the notes stay the same. Hold `SAVE` to undo the last change.

7 · Notes, keys and scales

You don't need to read music to play the FM-1. A few facts about the keyboard go a long way.

The keyboard

The keys repeat a pattern of 7 white and 5 black keys. Each group of 12 is an **octave**: the same note names again, twice as high. The white keys are named A to G; the white key just left of each pair of two black keys is **C**. Each key is one **semitone** above the key to its left, black or white.

`OCT-` and `OCT+` move the whole keyboard down or up an octave.

As you play, the top of the screen names what you're playing: the note with its octave (**D4**), or the chord when you hold several keys (**Dm**, **A7**, **Gsus4**; **C/E** is C major with E as its lowest note). It is a quick way to learn the chords in chapter 23 by ear and by name. MENU › CONTROL › NOTE NAMES turns it off.

> Don't hold OCT− and OCT+ down together: that starts a countdown to **update mode**, which the installer uses. If UPDATE MODE IN appears on the screen, let go: it says UPDATE CANCELLED and nothing has changed. If you hold on until the screen says UBOOT, the FM-1 is waiting for new firmware: connect it to the computer and run the installer again.

Scales and keys

A **scale** is a choice of 7 of the 12 notes that sound good together. A piece "in D minor" uses mostly the notes of the D minor scale, starts and ends on D, and feels like home on D. Minor scales sound sad, dark or ancient; that's why most dungeon synth is in minor.

The **root** is the home note (D in D minor). Every scale has a different mood:

Scale (SCL)	Mood
MIN (natural minor)	Sad, plain, the default of dungeon synth
DOR (Dorian)	Minor but hopeful; folk tunes, wandering
PHRY (Phrygian)	Dark and tense, the note above the root a little too close
HARM (harmonic minor)	Old-world, dramatic, "castle" sounding
PHDOM (Phrygian dominant)	Desert ruins and old temples: Phrygian with a bright major third
HUNG (Hungarian minor)	Gothic and haunted: harmonic minor with a raised fourth
DHARM (double harmonic)	Exotic and ceremonial: two wide steps, a sorcerer's tower
MPEN (minor pentatonic)	Five notes, nothing clashes: easy to improvise with
MAJ (major)	Bright, happy: a tavern, a sunrise

Let the FM-1 play only the right notes

On `SCL`, set **ROOT** (KNOB 1) and **SCL** (KNOB 2), then **QNT** (KNOB 3) to **WHITE**. Now the white keys play only the notes of your scale, starting on the root at the C key, and the black keys are silent. You can't hit a wrong note.

QNT **SNAP** keeps every key but moves a note that isn't in the scale down to the nearest one that is. QNT **OFF** is the plain keyboard.

> **TRY THIS · D minor on the white keys**
>
> 1. Stop playback. Choose T1 with `ALGORITHM`.
> 2. Tap `SCL`. Turn KNOB 1 to ROOT **D**, KNOB 2 to SCL **MIN**, KNOB 3 to QNT **WHITE**.
> 3. Play the white keys from C upwards: you hear the D minor scale. Play a slow, simple tune; finish it on the C key (now D) and hear it come home.
> 4. Change SCL to **PHRY** and play the same tune: darker. Try HARM: suddenly a castle.

The scale settings belong to the track, so set them on each track you play from the keys. Chords and the scale cheat sheet are in chapter 23.

8 · Tone: oscillators and the filter

The oscillator makes the raw sound; the filter carves it.

Waveforms

The oscillator repeats one shape, the **waveform**, over and over. Its shape decides the raw tone:

Wave	Sounds like
SIN (sine)	Pure and round, almost a whistle. No overtones.
TRI (triangle)	Soft and hollow: a recorder or a flute.
SQR (square)	Hollow and reedy: a clarinet, old video games.
SAW (sawtooth)	Bright and buzzy, full of overtones: strings, brass. The usual start for pads.
PWM (pulse width)	A square that keeps changing its shape: rich and moving.

Two oscillators slightly out of tune with each other (**detune**) beat against each other and sound wider and warmer, like several players at once.

The filter

The filter removes part of the sound. The usual one is a **low-pass filter**: it lets the low part through and takes away the bright part above a point called the **cutoff**.

- **CUT** (cutoff): turn it down and the sound gets darker, softer and further away; up, brighter and closer.
- **RES** (resonance): makes the filter ring at the cutoff, adding a whistling, vocal edge. High values sound "synthy".
- **KTR** (key tracking): how much the cutoff follows the notes, so high notes don't sound duller than low ones.

Not every engine is built this way. GURDY, VOICE, PHYS and the others have their own controls for tone (chapter 15), but the idea holds: some control makes the sound brighter or darker.

> **TRY THIS · Open the filter**
>
> 1. On T2, turn `PRESETS` to **CRYPT PAD** (an ANALOG sound) and go to `HOME`.
> 2. On HOME, KNOB 1 is CUT and KNOB 2 is RES. Hold a chord with one hand and slowly turn KNOB 1 down, then up: the pad closes and opens like a door.
> 3. Turn RES up halfway and sweep CUT again: hear the whistle ride along.
> 4. Tap `EDIT` for all eight ANALOG controls. Change WAVE (KNOB 1) between SAW, SQR, TRI and SIN and hear the raw tone change. Hold `SAVE` to undo if you get lost.

9 · Shape in time: the envelope

An envelope describes how a note changes from the moment you press the key until it falls silent.

The FM-1's envelope has four stages, called **ADSR**:

Stage	What it sets	Low	High
ATK attack	How long the note takes to reach full loudness	Instant: a pluck, a hit	A slow swell: strings, choirs
DEC decay	How long it takes to fall from the peak to the sustain level	A quick drop	A slow settle
SUS sustain	The loudness while the key stays held (a level, not a time)	Fades to silence: a harp, a bell	Holds: an organ, a pad
REL release	How long the note rings after you let go	Stops dead	Fades out slowly into the reverb

A **pluck** (harp, lute): attack 0, short decay, sustain 0. A **pad** (strings, choir): long attack, full sustain, long release. An **organ**: attack 0, sustain full, short release.

The envelope does more than loudness

Tap `ENV` again for **ENV DEST**: how much the same envelope also moves the filter (FLT), the pitch (PIT) and the engine's shape control (SHP). Envelope on the filter makes a note start bright and darken as it fades, as real plucked strings do.

> **TRY THIS · Turn a pad into a harp**
>
> 1. Keep CRYPT PAD on T2. On HOME, KNOB 3 is attack and KNOB 4 is release; or tap `ENV` for all four.
> 2. On ENV, turn ATK (KNOB 1) to zero, DEC (KNOB 2) to about a third, SUS (KNOB 3) to zero. Play a few notes: the pad is now a pluck.
> 3. Turn REL (KNOB 4) up and hear each pluck ring on after you let go.
> 4. Now the other way: ATK up high, SUS full. Each note swells in slowly, like a choir taking a breath.

10 · Movement: the LFO and modulation

A sound that never changes sounds like a machine. Modulation makes it move.

The LFO

An **LFO** (low-frequency oscillator) is an oscillator too slow to hear. Instead of making sound, it moves a control up and down for you, smoothly and endlessly.

- `LFO` page: **RATE** (how fast), **WAVE** (the shape of the movement: SIN and TRI glide smoothly, SQR jumps between two values, SAW ramps, S&H jumps to random values), **PHS** (where in its cycle each note starts) and **FADE** (how long it waits to come in after a note starts).
- `LFO` again for **LFO 2**: **SYNC** (OFF, or RATE locked to the tempo as a note length, from 4 bars to 1/32), **TRIG** (NOTE starts the LFO over with each note, FREE lets it run on) and **POL** (BI moves both ways, UNI only upward).
- `LFO` a third time for **LFO DEST**: how much it moves the pitch (PIT), the filter (FLT), the shape (SHP) and the loudness (AMP).

LFO to…	Is called	Sounds like
Pitch, slow and small	Vibrato	A singer or violinist holding a note
Loudness	Tremolo	A shimmering organ, a pulsing pad
Filter	A sweep, a "wah"	The sound breathing open and closed

The modulation matrix

Tap `LFO` a fourth time for **MOD**: four slots, each connecting a **source** to a **destination** with an amount (AMT, which can be negative to turn the movement upside down). KNOB 1 chooses the slot.

- **Sources:** LFO, ENV (the envelope), VEL (how hard a MIDI key is struck), KEY (how high the note is), RAND (a new random value each note), and from a MIDI controller MODW (the mod wheel), AT (aftertouch) and EXPR (an expression pedal).
- **Destinations:** pitch, cutoff, shape, loudness, pan, the four effect sends, the LFO's own rate, vibrato, and the engine's eight controls (E1–E8).

TRY THIS · Make it breathe

1. On T2 with CRYPT PAD, tap `LFO`: RATE to about a quarter, WAVE to TRI.
2. Tap `LFO` twice more (LFO DEST). Turn FLT (KNOB 2) up slowly while holding a chord: the pad starts to swell and fade by itself.
3. Set FLT back to 0 and turn PIT (KNOB 1) up only a little: vibrato. Turn RATE up: a nervous warble. Further: a broken tape.
4. On **MOD**, set slot 1 to SRC RAND, DST CUT, a small AMT: each note comes out a little different, as if played by a person.

11 · Voices: poly, mono and glide

How many notes a track plays at once, and how it moves between them.

Tap `EDIT` until the **VOICE** page shows. Its first knob, **VCE**, has four settings:

VCE	What it does	Good for
POLY	Several notes at once: chords	Pads, organs, harps
MONO	One note at a time; a new key cuts the old note	Basses, lead melodies, the hurdy-gurdy
LEG (legato)	One note; if keys overlap, the note slides on without starting again	Smooth, singing melodies; flutes and voices
UNI (unison)	One note played by several voices, detuned against each other	Huge, wide leads and drones

- **GLD** (glide, also called portamento): the pitch slides from one note to the next instead of jumping. A little sounds like a voice; a lot sounds like a siren.
- **GLMOD**: RATE (big jumps take longer to slide) or TIME (every slide takes the same time).
- **PRIO** (mono only): when several keys are held, which one plays: the LAST pressed, the LOW one or the HIGH one.
- **VOICE 2** (tap EDIT again): **DTUNE** (how far apart the unison voices are), **PAN** (where the track sits from left to right) and **MUTE**.

> The four tracks share 12 voices in all. A UNI track uses several voices for each note, so with long chords on other tracks, some notes may be cut short. If notes drop out, use MONO where a part only needs one note at a time.

> TRY THIS · A singing lead
>
> 1. Choose T1 (the hurdy-gurdy) and turn `PRESETS` to **LOFI FLUTE** or **TENOR RECORD**.
> 2. Tap `EDIT` until VOICE shows. Set VCE to **LEG** and GLD to about a fifth.
> 3. Play a slow tune and let the keys overlap a little: the notes slide into each other. Lift between notes and they start fresh.
> 4. Try VCE **UNI** on CRYPT PAD with one finger: one key, a wall of sound.

12 · Space: effects

Effects put the instrument in a room. In dungeon synth, the room is half the music.

Effect	What it does
Reverb	The echoes of a room, all blurred together: a closet, a chapel, a cathedral. GHOULBOX's HALL rings for up to 12 seconds.
Delay	Distinct repeats of what you play, each quieter than the last: a call across a valley.
Chorus	A slightly wavering copy of the sound mixed in, as if several players: wider and shimmering.
Distortion	Overdrives the sound until it breaks up: grit, growl.
Tape	The whole mix through an old cassette: wavering pitch, warmth, dull treble and hiss.
Crush	A cheap old sampler: rougher, grainier, robotic at the extreme.

Sends: one room, four players

There is one reverb, one delay and one chorus, shared by all four tracks, just as four musicians share one hall. Each track decides how much of itself to **send** into each: tap `FX`, and KNOB 1–4 are the track's distortion, chorus, delay and reverb amounts. A pad drowned in reverb and a dry, close melody in front of it is a classic dungeon-synth picture.

Tap `FX` again to reach the shared effects themselves: **SLICER**, **DLY**, **REVERB**, **CHORUS** and **TAPE** (chapter 16).

> **TRY THIS · From closet to cathedral**
>
> 1. Play the scene. Tap `FX` until **REVERB** shows. Turn SIZE (KNOB 2) all the way down, then slowly all the way up.
> 2. Turn DAMP (KNOB 3) up: the echoes go dark, like stone. Down: bright, like glass.
> 3. Tap FX again until **TAPE**. Turn TAPE (KNOB 1) up to 100 and listen to the pitch waver. Add CRSH (KNOB 2) slowly: grit, then crunch.
> 4. Back on the first **FX** page, choose T1 and turn its REV send (KNOB 4) to zero: the melody steps out of the hall and stands in front of you.

13 · Time: tempo, steps and the arpeggiator

Rhythm on the FM-1 comes from a clock that every track follows.

Tempo

The **tempo** is measured in **BPM**, beats per minute: 60 is one beat a second. Dungeon synth usually sits slow, 50 to 90. Turn `SELECT` to change it.

Steps

A **step sequencer** divides time into equal boxes, **steps**, and plays what is in each box in turn, round and round. Each track has its own row of up to 64 steps. Two settings decide how it moves; hold `SEQ` for them (KNOB 1 and 2):

- **LEN**: how many steps before it goes round again.
- **DIV**: how long each step is, as a fraction of a bar (a bar is four beats). 1/4 is one beat per step, 1/8 half a beat, 1/1 a whole bar. The power-on scene uses 1/8.

LEN 16 at 1/8 is two bars; LEN 4 at 1/1 is four bars, one chord each. Each track can have a different length, so parts drift against each other: a 7-step melody over an 8-step bass never repeats the same way twice in a row.

Swing (SWG) delays every second step a little: a lilt, a dance, a limp.

The arpeggiator

An **arpeggio** is a chord played one note at a time, as a harpist does. The arpeggiator does it for you: hold a chord, and it plays its notes in turn, in time with the tempo. Tap `ARP`:

- **MODE**: OFF, UP, DN (down), UPDN (up and down), RND (random), ORD (the order you pressed the keys), REPEAT (the whole chord again and again).
- **RATE**: how fast, as a note length. **OCT**: how many octaves it climbs (1–4). **GATE**: how long each note sounds, from short and plucked to tied together.
- **ARP 2**: SWG (swing), PROB (a chance each note plays, for a sparser, human pattern), **HOLD** (keeps playing after you let go: hands free) and ORD.

> **TRY THIS · A harp that plays itself**
>
> 1. Choose T1. Turn `PRESETS` to **FOLK HARP**. Set its scale to D minor on WHITE as in chapter 7.
> 2. Tap `ARP`: MODE **UPDN**, RATE **1/8**, OCT **2**.
> 3. Hold the C, E and G keys (a D minor chord in WHITE mode): the harp rolls up and down. Move the same three-finger shape to A, C, E (B♭ major) and back.
> 4. Tap ARP again and turn HOLD on: lift your hand and the harp keeps going. Turn `SELECT` and hear it follow the tempo.

PART III

Reference

Every page, sound and setting of GHOULBOX, for looking things up.

14. Every page

15. The sounds

16. Effects

17. Sequencer, patterns and songs

18. Saving your work

19. The menu

20. MIDI, USB and recording

14 · Every page

Each page button opens a group of pages; tap it again for the next (it remembers the last one you used). On every page, `KNOB 1-4` change the four values shown, left to right.

Button	Page	KNOB 1 · 2 · 3 · 4
`HOME`	HOME	The engine's four main controls (ANALOG: CUT · RES · ATK · REL)
	MIXER	The selected track's LEVEL · PAN · REV · MUTE (tap HOME on HOME)
	CLOCK	BPM · SWG · CLK (INT, USB, TRS)
	KEYS	MODE (OFF, LAYER, SPLIT) · TRK (the second track) · SPLT (where the keyboard divides) · OCT (chapter 15)
	DREAM	EVOL (OFF, ON) · DENS (1–8) · DRFT (1–8) · MOOD; a key composes a piece in its key (chapter 15)
`ENV`	ENV	ATK · DEC · SUS · REL
	ENV DEST	FLT · PIT · SHP (how much the envelope moves each)
`LFO`	LFO	RATE · WAVE · PHS · FADE
	LFO 2	SYNC · TRIG · POL
	LFO DEST	PIT · FLT · SHP · AMP
	MOD	slot (1–4) · SRC · DST · AMT
`FX`	FX	The track's sends: DST · CHO · DLY · REV
	SLICER	SLCR (OFF, GATE, STUT) · PAT (1–16) · RATE · DEPTH
	DLY	TIME · FDBK · COLR · MIX
	REVERB	TYPE (ROOM, SPRING, HALL) · SIZE · DAMP
	CHORUS	CRT (rate) · CDP (depth)
	TAPE	TAPE · CRSH · AMB · CRKL
`SCL`	SCL	ROOT · SCL · QNT (OFF, SNAP, WHITE, SEQ) · TRN (transpose)
	CHORD	CHRD (OFF, DIA3, DIA7, MAJ, MIN, DOM7, MAJ7, MIN7, SUS4, POW) · VOIC (CLOSE, OPEN, INV1, INV2, +OCT) · TEMP (EQUAL, PYTH, MEAN, JUST)

EDIT	EDIT 1 · EDIT 2	The engine's eight controls, four per page (chapter 15)
	RING	GURDY tracks only: RING (OFF, ON), the drones through the sequencer's rests (chapter 15)
	VOICE	VCE · GLD · GLMOD · PRIO
	VOICE 2	ALLOC · DTUNE · PAN · MUTE
GLO	SONG	SECTION (the row) · TRACK · PAT (A–D, or – for silent) · REPS (1–16)
SAVE	USER	slot (1–32) · LOAD · ERASE · SAVE
	PHRASES	The factory pattern to load
	PROJECT	SLOT (1–4) · – · LOAD · SAVE
	TOOLS	PAT (clear the track's steps) · SOUND (reset the sound) · ROW · SONG
	PRESETS	sound · engine · mark as favourite · LIST (ALL, FAV, or one kind: BASS, LEAD, PAD, PLUCK, KEYS, DRUM, FX, OTHER)
ARP	ARP	MODE · RATE · OCT · GATE
	ARP 2	SWG · PROB · HOLD · ORD
SEQ	STEP	STEP · NOTE · TIME (NOTE, TIE, REST) · FLAG (ACC, SLD, A+S); on a drum track STEP · LANE · HIT · ACC
	AUTOMATION	A list of the track's automation and its steps' CHANCE, RATCH and NUDGE: row · step · parameter · value (chapter 17)

Action knobs (LOAD, SAVE, CLEAR and the like): turn the knob right to pick the action, then press OCT+ to do it. Anything that would overwrite your work asks first.

> The PRESETS knob changes the sound on every page that is not a list, so you can audition sounds from anywhere. On the SEQ pages it moves the step cursor instead, and on the lists it moves the selection.

15 · The sounds

59 sounds across 13 engines. Hear them all at `ghoulbox.messy.quest/sounds`. GB marks GHOULBOX's own; the rest are Felucca's that fit the genre.

Engine	Sounds (in PRESETS order)
ANALOG virtual analog	SOFT PAD · PWM STR · SUB BASS · WIND · STRINGS · CRYPT PAD GB · FROST STR GB · WAR HORN GB · DIRGE BASS GB
FM6 6-operator FM	BELL · PAD · FRENCH HORN GB · VIRGINAL GB · GLASS CHOIR GB · DARK STRINGS GB · LOFI FLUTE GB · PIPE ORGAN GB · TUBA GB
PHASE phase distortion	STRING · GRIM BRASS GB
LOFI chiptune	TOWER LEAD GB · CASIO CHOIR GB · RECORDER GB
SAMPLE real recordings	BOWED PSALT GB · REN ORGAN GB · TENOR RECORD GB · FOLK HARP GB
VOICE sung vowels	CHOIR AAH · MONKS GB · CRYPT CHOIR GB · WRAITH GB · LOW DRONE GB
TRIO three oscillators	CHIP CHOIR · FANTASY PAD GB · MOURN HORN GB · STRING MACH GB · SHAWM GB
WHEEL tonewheel organ	FULL ORGAN · SOFT FLUTE · CRYPT ORGAN GB · CATHEDRAL GB · HARMONIUM GB · TOWER FLUTE GB
GRAIN granular	CLOUD PAD · FROZEN
PHYS physical models	LUTE GB · HARPSICHORD GB · DUNGEON HARP GB · GURDY DRONE GB
GURDY hurdy-gurdy	HURDY GURDY GB · DRONE WHEEL GB · DANCE GURDY GB · TROMPETTE GB · VIELLE GB · BAGPIPE GB
NOISE noise and ambience	CAVE WIND GB · TORCH GB
DRUM	CRYPT KIT GB · TOMB DRUMS GB

To jump between engines quickly: SAVE › PRESETS, KNOB 2. To mark a sound as a favourite, KNOB 3 on the same page; KNOB 4 then shows favourites only. SAVE › TOOLS › SOUND resets the track's sound to its engine's first preset (the notes stay).

The engines

Engine	How it makes sound	Controls to try first (EDIT)
ANALOG	Two oscillators and noise into a resonant low-pass filter, like the classic synthesizers. Pads, strings, horns, basses.	WAVE, DTN (detune), CUT, RES, DRV (drive)
FM6	Frequency modulation with six operators, like the famous 1980s keyboards: oscillators bending each other's pitch for bells, organs, brass, glassy tones.	SLOT (the patch), FB (feedback: grit), MLVL (brightness)
PHASE	Phase distortion: a smooth wave bent into brighter shapes.	WAVE, DCW (how far it's bent)
LOFI	Old game-console chips: pulse, triangle, noise. Chiptune melodies and 8-bit dungeons.	CHIP, WAVE, DUTY, VIB (vibrato)
SAMPLE	Recordings of real instruments (below).	SET (the instrument), BITS, CUT
VOICE	A sung tone shaped into vowels: monks, choirs, low chants.	VOWL, VOWL2 (the vowels), BRTH (breath)
TRIO	Three oscillators tuned in intervals, into a filter with several modes. Reeds, shawms, organ-like stacks.	INT2, INT3 (intervals), CUT, RES
WHEEL	A tonewheel organ with drawbar registrations and a rotating speaker.	REG (registration), PERC, ROTR (rotary)
GRAIN	Granular: thousands of tiny slices of a recording, smeared into clouds and textures.	POS, SIZE, DENS (density), SPRD
PHYS	Physical models: strings, bars and drums, simulated as they vibrate. Lutes, bells, dulcimers.	MODEL, BRIT (brightness), DAMP
GURDY	A hurdy-gurdy: a melody string over drones, with a buzzing bridge. GHOULBOX's own engine (below).	DRN, BUZZ, COUP, WHL
NOISE	Noise in many colours: wind, rain, crackle, a cave, a fire.	MODE, COLR, FREQ, RES
DRUM	Synthesized drums on eight lanes (below).	KIT, TUNE, DECY (decay)

On HOME, the four knobs are each engine's most useful four; EDIT shows all eight. To change a track's engine directly, hold EDIT and press a white key.

Real recordings (SAMPLE)

The SAMPLE engine plays four real instruments, recorded by Versilian Studios and released into the public domain (CC0): a **bowed psaltery** (a medieval zither played with a bow), a **Renaissance chamber organ**, a **baroque tenor recorder** and a **folk harp**. Held notes on the first three sustain for as long as you hold the key. SAMPLE's SET knob lists them first, then your own samples (USR1–USR3, uploaded or recorded in the web editor). An empty user slot plays a plain tone and the screen says NO SAMPLE.

The hurdy-gurdy (GURDY)

A hurdy-gurdy is a medieval string instrument played by turning a crank: a rosined wheel rubs the strings, keys on the side stop the melody string, and other strings drone on without stopping. One of them, the **trompette**, sits on a loose bridge that buzzes when the player jerks the crank, giving the dance its rhythm. GURDY plays one melody note at a time over its drones. Its eight controls (`EDIT`):

Control	What it does
`DRN`	The drone's note (C–B): set it to your song's key (D for D minor)
`DLVL`	Drone level
`5TH`	A second drone a fifth above, and its level
`WHL`	Wheel pressure: darker when low, brighter and rougher when high
`BUZZ`	The trompette: how hard it rattles (0: no buzz)
`COUP`	When it buzzes: always (HOLD), in rhythmic strokes in time (1/4, 1/8, 1/8T, 1/16), or by your own hand (CRANK, below)
`BODY`	The wooden body's resonance
`WOBL`	The wheel's unevenness: a slow wander in pitch and loudness

Turn the crank yourself

Set COUP to **CRANK** and the `SELECT` knob becomes the hurdy-gurdy's crank. Hold a key and turn: the faster you turn, the louder and brighter the strings. Stop turning and they fall silent, even with the key held; turn again and the note comes back. A quick flick of the wrist, a sudden speed-up, kicks the trompette's buzz, just as a player's strokes do (BUZZ sets how hard). WHL becomes **GEAR**: how fast you must turn for full sound (higher: less turning). While a track is in CRANK, change the tempo with `GLO` + SELECT or HOME › CLOCK. The sequencer still plays the notes, but they only sound while you crank: the pattern fingers the keys, you work the wheel.

Drones through the rests (RING)

On a real hurdy-gurdy the drones sound as long as the wheel turns, even when no key is pressed. Tap `EDIT` past the DRONE and WHEEL pages to **RING**. With RING **ON**, when the sequencer reaches a rest the melody string fades away (as fast as the sound's release) but the drones keep sounding until the next note. They stop when you stop the sequencer, or turn RING off. HURDY GURDY, DRONE WHEEL and BAGPIPE come with RING on; the other GURDY sounds, and songs saved before GHOULBOX 2.3, have it off and play as they always did.

Hands-free drones (LATCH)

Hold `GLO` and press the black key for a track one octave above its mute key (F♯, G♯, A♯ or C♯ in the upper octave, for tracks 1–4): the track is **latched**, and its key and its LTCH cell on the map light up. Now play a chord on it and let go: the chord keeps sounding. Select another track and play a melody over it, or crank the hurdy-gurdy with both hands free. Play a new chord to replace the drone; press one of the drone's notes to stop it (that press stays silent). Press the GLO key again to unlatch and end the drone. Other tracks never take the drone's voices, however busy they get (up to 8 latched notes). The latch isn't saved with the project, and loading a sound ends the drone. It does nothing on a DRUM track, and with the ARP on, ARP › HOLD does the same job. Recording writes the notes as long as you held the keys.

Two sounds at once (KEYS: LAYER and SPLIT)

Tap `HOME` on the home screen until **KEYS** shows (after MIXER and CLOCK). Turn KNOB 1 (MODE):

- **LAYER**: every key plays the selected track and a second track together: a recorder doubled by strings, a pad under a melody.
- **SPLIT**: the keys below a point play the second track, the rest the selected one: a drone or a bass in your left hand, the tune in your right.

KNOB 2 (TRK) picks the second track (T4 to begin with). KNOB 3 (SPLT) is the first key of the upper part (C4 to begin with: the seven keys below it play TRK). KNOB 4 (OCT) moves the second track's notes up or down by up to two octaves; a DRUM track keeps its drums where they are. Each track keeps its own sound, scale and chord settings, and each records what it plays when it is armed. KEYS is saved with the project. On the SEQ pages the keys play the selected track only, because there they write its steps.

Let it dream (DREAM)

Tap `HOME` on the home screen until **DREAM** shows (after KEYS). Press a key: the FM-1 composes a slow piece in that key for your four tracks and starts playing it. Each track gets a part that suits its sound: a drone holds the key note, a bass walks the chords' roots, a pad or a choir holds the chords, a harp or a lute breaks them into notes, a melodic sound sings a short phrase and answers it, a second melodic sound adds a few high notes, drums play a slow ritual. Choose the sounds first, then dream: a different set of sounds makes a different piece.

- **EVOL** (KNOB 1): ON, and the piece keeps changing by itself while it plays: a new chord, a new phrase, a part played anew, a part resting or coming back. The drone and the opening phrase stay, so it still sounds like the same piece.

- **DENS** (KNOB 2): how much each part plays, 1 (sparse) to 8 (busy). The notes have a chance each, so even without EVOL no two passes are quite the same.
- **DRFT** (KNOB 3): how often EVOL changes something: 1 every eight passes, 8 every pass (a pass is about 27 seconds at 72 BPM).
- **MOOD** (KNOB 4): the scale: MIN (natural minor), DOR (Dorian), PHRY (Phrygian), HARM (harmonic minor), MAJ, MIX (Mixolydian). Set it before you press the key.

DREAM writes ordinary patterns, 64 steps at 1/8, and sets each track's ROOT and SCL to the key, so you can play along in key, edit the steps, mute parts, and save the project like any other. **It replaces all four tracks' patterns**: the first time, if there are steps to lose, it asks (OCT+ yes, OCT− no). Change the tempo with `GLO` + SELECT as usual.

Drums (DRUM)

The DRUM engine's **CRYPT** kit is built for dungeon synth: a low **frame drum**, a **rope snare**, a hand **slap**, a **shaker**, **jingles**, a **timpani**, a wooden **stick** and a **gong**. Two sounds use it: **CRYPT KIT** and **TOMB DRUMS** (lower, longer and further back in the hall), both suggesting the TOMBBEAT pattern. KIT also offers Felucca's kits (STD, 80, 10, 66, 55, 77) if you want something less ancient.

The keys follow the General MIDI drum map, the same one drum machines and computer drum tracks use: C is the frame drum, D the rope snare, A the timpani. On a DRUM track, `SEQ` › STEP becomes a **drum grid**: the white keys are the steps of the current bar and the black keys choose the lane (the drum); KNOB 1–4 are STEP, LANE, HIT and ACC (accent).

16 · Effects

Each track sends to shared effects; the amounts are on the `FX` page (KNOB 1–4: distortion, chorus, delay, reverb). Tap `FX` again to reach the effects themselves. Distortion and the slicer belong to each track; the others are shared.

REVERB

- **TYPE** (KNOB 1): ROOM, SPRING or **HALL**. GHOULBOX starts in HALL.
- **SIZE**: from a 1.5-second stone chamber to a 12-second cathedral.
- **DAMP**: darkens the tail.

TAPE

- **TAPE**: the whole mix through a worn cassette: wow and flutter (wavering pitch), saturation, treble loss and hiss. GHOULBOX starts at 40.
- **CRSH**: a cheap sampler: fewer bits and a lower sample rate. Off by default; a little adds grit, a lot turns to crunch.
- **AMB**: the ambience, a low cave wind that rises and falls, in the same reverb as your tracks. It uses no track and no voice, and plays even when the song is stopped, like the air in a room. Off (0) by default; the first half of the knob is subtle.
- **CRKL**: the torch crackle in the ambience: from 0 (wind only) to a busy fire. AMB sets the level of both.

DLY

TIME (as a note length, so it follows the tempo), **FDBK** (feedback: how many repeats; past 100 it builds up on itself, so ride it carefully), **COLR** (darker or brighter repeats) and **MIX**. With HALL selected, the delay is limited to 0.74 s, because the hall borrows part of its memory.

CHORUS

CRT (rate) and **CDP** (depth). Slow and deep for a seasick, warped-tape shimmer; fast and shallow for a thicker sound.

SLICER

Chops the track's sound in time with the tempo. **GATE** cuts it on and off in one of 16 rhythm patterns (a pulsing pad); **STUT** repeats short slices of it (a stutter). RATE sets how fine the chopping is, DEPTH how deep.

The FX layer

Hold `FX` to play performance effects from the keys (repeat, reverse, filter sweeps, tape stop, freeze, a harmonizer with octave shimmer) and mute tracks with the black keys. With MENU › FX LATCH on, they toggle instead of needing to be held. The screen shows which key does what.

17 · Sequencer, patterns and songs

Each track has its own sequence of up to 64 steps, with chords, ties, accents, slides, per-step chance and ratchets.

Load a factory pattern

`SAVE` › PHRASES (tap SAVE twice) lists ready-made patterns. Choose one with KNOB 1 (or PRESETS) and press `OCT+` to load it into the selected track; if the track has notes you wrote, it asks first. GHOULBOX adds nine, all in D minor:

Pattern	What it is	Try it on
DIRGE	Slow pad chords	CRYPT PAD, DARK STRINGS, PIPE ORGAN
BALLAD	A melody	HURDY GURDY, LOFI FLUTE, BAGPIPE
CHANT	Low voices	MONKS, LOW DRONE
PEDAL	One held low D	DRONE WHEEL, any bass
MARCH	A stepping bass	DIRGE BASS, TUBA
LAMENT	A falling melody	TENOR RECORD, BOWED PSALT
ARPEGGIO	Broken chords	FOLK HARP, VIRGINAL, a lute
ANTIPHON	A call answered an octave lower	Two voices, a gurdy
TOMBBEAT	A slow beat for the CRYPT drums	CRYPT KIT, TOMB DRUMS

Patterns 1–13 are Felucca's. Loading a sound never changes your pattern, and loading a pattern never changes your sound.

Write notes step by step

1. Go to `SEQ` › STEP. The cursor is the step being edited (KNOB 1 or PRESETS moves it).
2. Press a key: it is written into the step. Press several together for a chord (up to 4 notes on a POLY track). When you let go of all keys, the cursor moves to the next step.
3. Tap `EDIT` to empty the step and move on: a rest.
4. Set TIME (KNOB 3) to **TIE** to hold the previous note through this step instead of playing a new one.
5. FLAG (KNOB 4): **ACC** plays the step louder, **SLD** slides into the next note, **A+S** both.

Record live

Press `REC` to arm the selected track (from any page), press `PLAY`, and play: what you play is written into the steps as they pass. The sequence loops, and you can keep adding on each pass (overdub). Press REC again to stop recording.

Knob moves on an armed track are recorded too (**automation**): the sound changes at those steps every time round. `SEQ` › AUTOMATION (tap SEQ twice) lists them; its first row turns them off or clears them.

Chance and ratchets

A step can have a **CHANCE** (the percentage of times it plays: lower values make a pattern sparse and alive) and a **RATCH** count (the step played 2, 3 or 4 times within its length: a drum roll). Add them on `SEQ` › AUTOMATION: turn KNOB 1 to the last row, **+ ADD**; choose the step with KNOB 2 and CHANCE or RATCH with KNOB 3, and press `OCT+` (it starts at 50 % or ×2). KNOB 4 then sets the value. `EDIT` on a row puts the step back to 100 % or ×1.

Length, speed and clearing

- Hold `SEQ`: KNOB 1–4 are LEN (1–64 steps), DIV (step length, from 4 bars to 1/32), SWG (swing) and GATE (how long notes sound within their step).
- **Clear** the track's steps: hold `REC` (or `SEQ`) and press the lowest F key, CLEAR. It clears at once; holding `SAVE` undoes it. `SAVE` › TOOLS › PAT does the same but asks first.

Songs

A song chains up to 16 rows (sections) into a longer piece. Its patterns are **saved projects**: **A** is the steps of project 1, **B** of project 2, **C** of 3 and **D** of 4. Each row says, for each track, which of them it plays (or **–**, silent), so in one section the drone can keep A while the melody moves on to B. The song plays them with the sounds currently loaded.

1. Make a section and save it as project 1 (chapter 18). Change the notes for the next section and save it as project 2, and so on.
2. Tap `GLO` for the SONG page. KNOB 1 picks a row. The last row, **+ ADD**, makes a new one when you set a pattern there (a new song starts with only + ADD): a copy of the row before, or every track on that pattern for the first. KNOB 2 picks a track and KNOB 3 its pattern (A–D or –); the A, B, C and D keys (the lowest A, B, C and D) set it too. KNOB 4 sets how many times the row plays (1–16).
3. Press `PLAY` on the SONG page to play the song from the top. Edit the sections while stopped.

For example, rows A ×2, B ×1, A ×2, C ×1 give a verse, a bridge, the verse again and an ending (each with all four tracks on that letter). `SAVE` › TOOLS › ROW deletes a row and SONG clears the whole song.

18 · Saving your work

The FM-1 keeps two kinds of things: **sounds** (one track's sound) and **projects** (everything). Saving works only while stopped.

Save a project (4 slots)

A project keeps the whole session: all four tracks, their sounds and sequences, songs, tempo and effects.

1. Stop. Tap `SAVE` until PROJECT shows.
2. Choose the slot with KNOB 1. Turn KNOB 4 (SAVE) right to pick it, and press `OCT+`. If the slot already holds a project, it asks first: OCT+ again.
3. Name it with the keys, and press `OCT+` to save.

To load: same page, KNOB 3 (LOAD) right, then `OCT+`.

Save a sound (32 user slots)

1. Stop and tap `SAVE`: it opens on USER.
2. Pick a slot with KNOB 1, press `OCT+`, name it with the keys, and press `OCT+` again.

Your sounds appear after the factory sounds when you turn `PRESETS`. On USER and PROJECT, `EDIT` renames the selected slot.

Undo

Hold `SAVE` to go back before the last sound load, pattern load or clear (hold again to redo).

Backups

The web editor (on the installer site) makes a full backup of the FM-1 to your computer and restores it. Do one now and then, and before any experiment you might regret.

> The FM-1 comes back on with your last session: it saves it once you have been stopped for about 10 seconds (MENU › SYSTEM › RESTORE LAST, ON since 2.1). Set RESTORE LAST to OFF to start with the power-on scene (MENU › SYSTEM › SCENE) every time instead. Your saved projects stay in their slots; load one from `SAVE` › PROJECT.

19 · The menu

Hold `HOME` for the menu. `ALGORITHM` moves between its five tabs, `PRESETS` through the rows, and any knob or `OCT-` / `OCT+` changes a value. Press `HOME` to close it.

Tab	Settings
DISPLAY	**COLOR** (palette; GHOULBOX's is CRYPT) · **STYLE** (FLAT or LINE) · **LARGE** (bigger knob cards) · **ANIM** (rolling digits ON or OFF; IDLE also plays a light show on the keys when you leave it alone) · **LEDS** (OFF, DIM LO, DIM HI, INV) · **SCREEN OFF** (after NEVER, 5, 15, 30 or 60 minutes) · **SCOPE** (OUT draws the sound after MASTER, MIX before it) · **HOME** (SCOPE, or TRACKS: all four tracks at a glance)
CONTROL	**HOLD** (how long a quick layer takes to open) · **KNOB ACCEL** (fast turns move further) · **FX LATCH** · **BPM LOCK** (SELECT stops changing the tempo) · **SCALE LEDS** (the keys light the scale) · **STEP PREVIEW** (on STEP, the cursor's step sounds as you move to it) · **CHORD ENTRY** (HOLD writes the keys held together into a step; ADD adds each key to it) · **NOTE NAMES** (the note or chord you play, named at the top of the screen)
AUDIO	**SPEAKER EQ** (FLAT, LOWCUT, BASS+; keep FLAT when recording) · **USB LEVEL** (MASTER or FIXED) · **CLICK** (a metronome: OFF, REC while recording, or ON) · **CLICK LEVEL** · **COUNT-IN** (OFF, 1 BAR, 2 BARS before recording) · **TUNE** (±50 cents, saved with the project)
MIDI	**MIDI IN** (which channels play which track: chapter 20; saved with the project)
SYSTEM	**SCENE** (what plays at power-on: DUNGEON, CRYPT, TAVERN, CHAPEL or BLANK; chapter 3) · **USB SERIAL** (OFF from 1.2 on; turn it ON only for Felucca's developer console, and keep it OFF on macOS 13–15 so the USB audio input appears) · **RESTORE LAST** (ON, the default: your last session comes back at power-on; OFF: the scene) · HARDWARE CALIBRATION · **INFO** (the firmware, the USB connection and the CPU load) · ABOUT (the version)

20 · MIDI, USB and recording

MIDI is the language electronic instruments use to talk to each other: which key went down, how hard, which knob moved. It carries no sound, only instructions.

What can talk to the FM-1

- **A computer** over USB: a DAW (music software) can play the FM-1's tracks, and the FM-1's keys send notes back to it. No driver needed.
- **A MIDI keyboard or controller**: over USB through a computer, or into the MIDI IN socket with a 3.5 mm adapter (chapter 1). A bigger keyboard gives you more octaves and touch-sensitive keys.
- **Another instrument's clock**: `HOME` › CLOCK › CLK (tap HOME twice on HOME) set to USB or TRS makes the FM-1 follow the tempo and start/stop of a computer or drum machine.

Channels

MIDI has 16 channels, like 16 radio stations on one cable. With MENU › MIDI › MIDI IN on **CH1-4**, channels 1–4 play tracks 1–4 and the rest are ignored: a DAW can play all four tracks at once. **CH5-8**, **CH9-12** and **CH13-16** do the same with those channels, to share a cable with other instruments. With **SEL**, every channel plays the selected track. Pitch bend and the sustain pedal work too.

Control changes (CC)

Knobs on a MIDI controller send **control changes**, numbered 0–127. These turn the FM-1's controls: those marked *track* on the channel's track, the others for the whole mix:

CC	Control
7 · 10	Track level · pan (track)
73 · 75 · 79 · 72	Attack · decay · sustain · release (track)
91 · 93 · 94 · 92	Reverb · chorus · delay · distortion sends (track)
76 · 77 · 78	LFO rate · LFO to pitch (vibrato) · LFO fade-in (track)
110 · 111	LFO to filter · LFO to loudness (tremolo) (track)
21–24	The four engine knobs (the HOME page's) (track)
5 · 71	Glide · the engine's resonance, where it has one (track)
85 · 86	CRSH · TAPE
102 · 103	REVERB SIZE · DAMP

104 · 105 · 106 · 107	DLY TIME (from 4 bars at 0 to 1/32 at 127) · FDBK · COLR · MIX
108 · 109	CHORUS rate · depth
74	The FX layer's filter on the whole mix: below 64 low-pass, above high-pass, 64 off (like the FX layer's own filter, it resets when the FX layer is let go or the menu opens)

On the selected track, while it is armed (`REC`), these moves are recorded as automation, as when turning its knob. The mod wheel, aftertouch and expression pedal are sources in the MOD matrix (chapter 10).

Recording to a computer

The FM-1 appears on a computer as a USB audio input named **"Felucca"**, carrying the master output in stereo at 44.1 kHz, with no driver. Chapter 25 walks through it.

PART IV

Making music

The genre in a page, a whole piece built step by step, the chords you need, recipes, and how to get it out of the box and into the world.

21. What dungeon synth is
22. Your first piece, step by step
23. Minor keys and chords
24. Dungeon recipes
25. Recording and sharing

21 · What dungeon synth is

Slow keyboard music for imagined castles, forests and crypts, usually made by one person at home.

Dungeon synth grew out of the early-1990s black-metal scene, from the synthesizer intros and side projects of musicians such as Mortiis, and was given its name later. It sounds medieval or fantastical rather than modern: drones, simple modal melodies, choirs, horns, harpsichords, all drowned in reverb, often on cheap keyboards and recorded onto cassette. The roughness is part of the charm.

What makes it sound like dungeon synth

- **Slow.** 50–90 BPM, long notes, room to breathe.
- **Minor and modal.** Natural minor, Dorian, Phrygian, harmonic minor (chapter 23).
- **Drones.** One low note held under everything: the GURDY drones, PEDAL, LOW DRONE.
- **Few, simple parts.** A pad, a melody, a bass or drone, perhaps a drum. Repetition is a feature: let a phrase come round many times.
- **Huge reverb.** HALL with a big SIZE. The space is an instrument.
- **Lo-fi.** TAPE wobble and hiss, a little CRSH. It should sound found, not produced.
- **Old-world instruments.** Organs, harps, recorders, horns, choirs, bells, the hurdy-gurdy: the sounds GHOULBOX is built around.

Its neighbours

Dungeon synth borders **dark ambient** (no melody, only atmosphere), **winter synth** (colder, more echo), **comfy synth** (cosy, major-key, pastoral) and old **video-game music** (try LOFI). Any of them can be made on GHOULBOX.

> The way in is to listen. Pick an album you love, play along with it on the FM-1 in WHITE mode, and notice how few notes it really uses.

22 · Your first piece, step by step

A complete piece in four parts and two sections, from empty tracks to a song. Take your time: it's an hour's work the first time, and ten minutes after that.

1. Clear the table

1. This walk-through starts from the **DUNGEON** scene, the one a new FM-1 plays. If yours comes back with something else, hold HOME and set SYSTEM › SCENE to DUNGEON and RESTORE LAST to OFF, switch off and on, then set RESTORE LAST back to ON. Don't press PLAY yet.
2. For each track, T1–T4 (`ALGORITHM`): hold `REC` and press the lowest F key (CLEAR). The steps are cleared; the sounds stay.
3. Turn `SELECT` to about 66 BPM.

2. The drone (track 4)

1. Choose T4. Its sound is DRONE WHEEL, a hurdy-gurdy drone. On `EDIT`, check DRN is **D**.
2. Tap `SAVE` twice for PHRASES, choose **PEDAL** and press `OCT+`. Press PLAY: a low D that never stops. Leave it playing.

3. The chords (track 2)

1. Choose T2 (CRYPT PAD). On `SCL`: ROOT **D**, SCL **MIN**, QNT **WHITE**.
2. Hold `SEQ` and set LEN (KNOB 1) to **4** and DIV (KNOB 2) to **1/1**. Four steps, a bar each.
3. `SEQ` › STEP, cursor on step 1 (KNOB 1). Press and release each chord shape in turn; each fills one step:
   - C + E + G (D minor)
   - A + C + E (B♭ major)
   - B + D + F (C major)
   - C + E + G (D minor again)
4. The pad now plays **Dm – B♭ – C – Dm**, one of the oldest progressions in the genre, over the drone.

4. The melody (track 1)

1. Choose T1 (HURDY GURDY). On `SCL`: ROOT D, SCL MIN, QNT WHITE. On `EDIT`, BUZZ to 0 for now: a plain, sad gurdy.
2. Hold `SEQ`: LEN **32**, DIV **1/8** (four bars, the same length as the chords).
3. Play along with the chords, slowly, until a few notes feel right. Simple is good: a handful of notes, mostly stepping to neighbours, ending on the C key (D).

4. Press `REC`, wait for the loop to come round, and play it. Press REC again when you're done. Not right? Hold `REC` and press the lowest F key to clear, and try again.

Rather not play it live? Load **LAMENT** or **BALLAD** from PHRASES instead, or write it a note at a time on STEP (chapter 17).

5. The drum (track 3)

1. Choose T3. Turn `PRESETS` to **TOMB DRUMS**. Load **TOMBBEAT** from PHRASES.
2. Too busy? On `SEQ` › AUTOMATION, add a CHANCE of 50% to a few steps (chapter 17): the beat thins out and never repeats exactly.

6. Mix and room

1. Hold `GLO` and turn KNOB 4 and KNOB 3 (the drone's and the drum's levels) down until the melody and chords sit on top.
2. `FX`: give the pad and drum plenty of REV, the melody a little less. `FX` › REVERB: SIZE high. `FX` › TAPE: TAPE 60.

7. Save it, then make section B

1. Stop. Save as **project 1** (chapter 18). Call it VERSE.
2. On T2, go to STEP, move the cursor back to step 1 (KNOB 1) and play new chords over the old ones: **Gm – Dm – A – A**: keys F + A + C, C + E + G, and for A major, set SCL to HARM first, then G + B + D twice. (Harmonic minor turns the fifth chord major: drama.)
3. On T1, turn BUZZ up and COUP to **1/8**, and record a new melody, or mute it for a quieter middle.
4. Save as **project 2**, BRIDGE.

8. The song

1. Tap `GLO` for the SONG page. It shows only **+ ADD**. Press the **A** key: the first row, every track on A. Set its REPS (KNOB 4) to 2.
2. Turn KNOB 1 to **+ ADD** and press the **B** key: a second row. Its other tracks are still on A: for each one (KNOB 2: T1 to T4) press **B**. REPS 1.
3. On **+ ADD** again, press **A** and set every track back to **A** the same way; REPS 2. The rows read **A ×2**, **B ×1**, **A ×2**.
4. Press `PLAY` on the SONG page. That's a piece.

> The song plays the steps of each section with the sounds loaded right now, so the BUZZ you added in section B stays on when A comes back. Use the tracks' levels, or a track set to – on a row, rather than sound changes to make sections differ, or set the sound back before saving each section.

23 · Minor keys and chords

A cheat sheet. You need no theory beyond this page to write dungeon synth.

Chords in a nutshell

A **chord** is three or more notes played together. The basic chord, a **triad**, is built by taking every other note of the scale: the 1st, 3rd and 5th; or the 2nd, 4th and 6th; and so on. **Major** chords sound bright, **minor** chords sad.

Shapes on the white keys

With QNT on **WHITE**, every-other-white-key is always a chord of your scale. The shapes below give the chords of D minor with ROOT D and SCL MIN; change ROOT and the same shapes play the same chords in the new key.

Keys to press	In D minor	Degree	Feel
C · E · G	**D minor** (D F A)	i	Home
D · F · A	E diminished (E G B♭)	ii°	Tense, unstable: use sparingly
E · G · B	**F major** (F A C)	III	Hope, light
F · A · C	**G minor** (G B♭ D)	iv	Deeper sadness
G · B · D	A minor (A C E)	v	Restless, leading home
A · C · E	**B♭ major** (B♭ D F)	VI	Grand, epic
B · D · F	**C major** (C E G)	VII	Marching, heroic

With SCL **HARM** (harmonic minor), G · B · D becomes **A major** (A C♯ E): a strong pull back home. With **PHRY**, D · F · A becomes **E♭ major**: ominous.

Progressions that work

Progression	In D minor	Keys	Character
i – VI – VII – i	Dm B♭ C Dm	CEG · ACE · BDF · CEG	The classic: a quest
i – iv – i	Dm Gm Dm	CEG · FAC · CEG	A slow lament
i – VII – VI – VII	Dm C B♭ C	CEG · BDF · ACE · BDF	Descending, endless
i – III – VII – i	Dm F C Dm	CEG · EGB · BDF · CEG	Brighter, a sunrise over ruins
i – V (HARM)	Dm A	CEG · GBD	Castle drama
i – ♭II (PHRY)	Dm E♭	CEG · DFA	Dread

One finger, whole chords

On `SCL` › CHORD, set CHRD to **DIA3**: each key plays the triad of the scale that starts on it (DIA7 adds a fourth note for a richer sound). In WHITE mode with ROOT D, one finger on the C key plays D minor, on A plays B♭, and so on. VOIC spreads the notes: OPEN for a wide, airy pad, +OCT to double the bass an octave down.

Old tunings

A modern keyboard is tuned in **equal temperament**: every step the same size, so every key sounds equally (slightly) out of tune. Medieval and Renaissance players tuned to pure ratios instead, and their chords shimmer less and ring more. On `SCL` › CHORD, **TEMP** (KNOB 3) tunes the track from its ROOT:

TEMP	Sound
EQUAL	The default: every note as on any synth
PYTH (Pythagorean)	Pure fifths, sharp bright thirds: drones, plainchant, open fifths
MEAN (meantone)	Pure major thirds, soft and sweet: a chapel organ, slow chords
JUST (just intonation)	Pure chords: in a minor key, i, iv and ♭VI ring still

Set ROOT to your key first: the tuning is built from it. Chords on far-away notes (the sharp fourth, the flat second) can sound sour; that is the price of the pure ones. GURDY's drones follow TEMP too, so a hurdy-gurdy in PYTH buzzes a pure fifth.

Melody tips

- Start and end on the root. Lean on the notes of the current chord on its strong beats.
- Move mostly by step (to neighbouring white keys); save leaps for moments.
- Repeat a phrase, then change only its ending.
- Let the melody rest. Silence in a big reverb is music too.

24 · Dungeon recipes

Starting points. Load the sound with PRESETS, then adjust.

Old Tower-style pad

CRYPT PAD or DARK STRINGS on a track, the DIRGE pattern from SAVE › PHRASES, REVERB SIZE high, TAPE around 60. Slow everything down: tempo 60–70, DIV 1/4 or 1/2.

Lone hurdy-gurdy

HURDY GURDY with DRN set to your key. Use COUP 1/8 for rhythmic buzz strokes on a dance tune, or BUZZ at zero for a sad, plain drone.

Monastery

MONKS on one track and CRYPT CHOIR an octave above on another, the CHANT pattern, REVERB HALL with SIZE at maximum. Add PIPE ORGAN or REN ORGAN holding the root.

Tavern ballad

FOLK HARP or a lute on an arpeggio (ARP), TENOR RECORD or LOFI FLUTE for the tune, BAGPIPE for a reedy lead over its drones. SCL DOR or MAJ for a lighter mood, a little swing.

War council

FRENCH HORN or WAR HORN playing slow open chords (VOIC OPEN), TUBA or DIRGE BASS on MARCH, TOMB DRUMS on TOMBBEAT. SCL HARM.

The cave

A NOISE sound for dripping water or wind on one track, LOW DRONE on another, nothing else. DLY with long TIME and high FDBK on the noise. Dark ambient in two tracks.

8-bit dungeon

LOFI sounds for melody and bass, ARP on UPDN at 1/16 for the chords, ROOM reverb instead of HALL, no tape: an old console's dungeon level.

Lo-fi cassette demo

Any of the above with TAPE 80–100 and CRSH 30–50: the sound of a 1990s dungeon-synth tape.

25 · Recording and sharing

The FM-1 is its own audio interface: one USB-C cable carries your music to a computer, with no extra hardware.

Into a computer

1. Hold `HOME` for the menu. On AUDIO, set SPEAKER EQ to **FLAT** (the other settings colour the recording). For a level that doesn't depend on the MASTER knob, set USB LEVEL to **FIXED**.
2. Connect the FM-1 with a USB data cable. It appears as an audio input named **"Felucca"**, stereo, 44.1 kHz. No driver is needed on Windows, macOS or Linux. (On a Mac with macOS 13–15, MENU › SYSTEM › USB SERIAL must be OFF; it is, from 1.2 on.)
3. In recording software (Audacity is free; any DAW works), choose "Felucca" as the input, stereo, and record while the song plays.
4. Watch the level meter: the loudest moments should stay below the top. If it hits the top, turn the track levels down (hold `GLO`: KNOB 1–4), or MASTER if USB LEVEL is on MASTER.

Into a recorder or a mixer

Use the 3.5 mm audio out with a stereo cable. Turn MASTER to about three-quarters and set the level on the receiving end.

Finishing

- Trim the silence at the start and end, and let the last reverb tail ring out fully before cutting.
- Fade in and out if the piece starts or ends abruptly.
- Export as WAV for keeping, MP3 for sharing.

Sharing

Dungeon synth lives on Bandcamp, YouTube, SoundCloud and small cassette labels. Tag it "dungeon synth" and the instrument (FM-1, GHOULBOX); people do look. If you'd like the GHOULBOX community to hear it, post a link in the project's GitHub discussions or wherever you found GHOULBOX.

PART V

Help

When something goes wrong, a word you don't know, and who made what.

26. Troubleshooting

27. Glossary

28. Credits and licence

26 · Troubleshooting

Problem	What to try
No sound	Turn up MASTER; check the track isn't muted (HOME › MIXER, or hold GLO: a muted track's black key is dark) and its level isn't at zero; headphones in the jack silence the speaker.
Sound from the speaker even with headphones in	Use a stereo (TRS) plug; a headset (TRRS) plug may not switch the speaker off.
Notes cut off or missing	All four tracks share 12 voices. Use MONO on parts that play one note at a time, fewer UNI tracks, or shorter releases.
A key plays nothing	QNT is on WHITE: the black keys are silent. Set QNT to OFF on `SCL` for the full keyboard.
The keys play the wrong notes	Check SCL (ROOT, QNT, TRN) and the octave (OCT− / OCT+).
"STOP TO EDIT" / "STOP TO SAVE"	Stop playback first. Songs and saving need the transport stopped.
The song won't play	Each row's section (A–D) needs a saved project in that slot (A = project 1). The SONG page needs at least one row.
The tempo won't change	MENU › CONTROL › BPM LOCK is on, or HOME › CLOCK › CLK follows an external clock (set it to INT).
The installer can't find the FM-1	Use a USB data cable, close other music software, use Chrome or Edge.
The installer stops after "update mode"	Unplug and replug the USB cable and press Install again. The FM-1 waits in update mode until finished.
"NO SAMPLE" and a plain tone	The SAMPLE or GRAIN source is an empty user slot (USR1–3). Pick a built-in set or upload a sample in the web editor.
A keyboard or DAW plays the wrong track	MENU › MIDI › MIDI IN: CH1-4 sends channels 1–4 to tracks 1–4 (CH5-8 and the rest: those channels); SEL sends every channel to the selected track.

A keyboard on MIDI IN plays nothing	Try the other type of 3.5 mm MIDI adapter (Type A or B), and check the keyboard sends on a channel MENU › MIDI › MIDI IN listens to (1–4 on CH1-4), or set MIDI IN to SEL.
No USB audio on a Mac	MENU › SYSTEM › USB SERIAL OFF (macOS 13–15).
The recording sounds thin or boomy	MENU › AUDIO › SPEAKER EQ to FLAT.
The screen is black after an interrupted update	If GHOULBOX itself won't start, switch the FM-1 on while holding `OCT-`: it starts in its USB rescue (no sound, no menu), and the installer can install again from there; two failed starts in a row open it on their own. If the computer sees a device named WL80UBOOT, the FM-1 is in its boot mode and can be recovered: try another cable and run the installer again. Felucca's README describes further recovery.
Something else	Report it at `github.com/jasonpersinger/ghoulbox-fm1-dungeon-synth/issues` with your OS, browser and the installer's log text.

27 · Glossary

Accent
A step played louder than the others (FLAG ACC).

ADSR
Attack, decay, sustain, release: the four stages of an envelope (chapter 9).

Arpeggiator
Plays a held chord one note at a time, in time with the tempo (ARP).

Attack
How long a note takes to reach full loudness.

Automation
Knob moves recorded into the sequence so they happen by themselves.

Bar
A group of four beats; music is counted in bars.

BPM
Beats per minute: the tempo.

Cent
A hundredth of a semitone; used for fine tuning (TUNE).

Chord
Three or more notes played together.

Chorus
An effect that adds a wavering copy of the sound, making it wider.

Cutoff
The point above which a low-pass filter removes the sound's brightness (CUT).

DAW
Digital audio workstation: music software for recording and arranging.

Decay
How long a note takes to fall from its peak to the sustain level.

Delay
An effect that repeats the sound, each repeat quieter.

Detune
Putting two oscillators or voices slightly out of tune with each other, for width.

Drone
A long, unchanging note held under the music.

Engine
One of GHOULBOX's 13 ways of making sound.

Envelope
A shape over time, started by each note, that moves loudness (and more).

Filter
Removes part of a sound's frequencies; changes its tone.

FM
Frequency modulation: oscillators changing each other's pitch very fast, making complex tones.

Formant
The resonances that make a voice sound like one vowel or another (VOICE).

Gate
How long a note sounds within its step.

Glide
A slide in pitch from one note to the next (also portamento).

Granular
Making sound from many tiny overlapping slices of a recording (GRAIN).

Legato
Notes joined smoothly, without starting again.

LFO
Low-frequency oscillator: a slow, inaudible wave that moves other controls.

MIDI
The language instruments use to send notes and control moves to each other.

MIDI CC
A control change: a MIDI message for a knob or slider, numbered 0–127.

Mode (scale)
One of the old scales: Dorian, Phrygian and others.

Modulation
One thing moving another: an LFO moving pitch, an envelope moving the filter.

Mono / poly
One note at a time / several notes at once.

Octave
The distance from a note to the next of the same name: twice the pitch.

Oscillator
The part that makes the raw sound by repeating a waveform.

Overdub
Recording more on top of what is already recorded.

Pad
A soft, sustained background sound, usually chords.

Pan
Where a sound sits between left and right.

Pattern
A track's row of steps.

Preset
A ready-made sound (also: patch).

Progression
A sequence of chords.

Project
A saved session: all four tracks, sounds, steps and settings.

Quantize
Moving notes onto a grid: of pitch (a scale) or of time (steps).

Ratchet
A step played several times quickly within its length.

Release
How long a note rings after the key is let go.

Resonance
A filter ringing at its cutoff, adding a whistling edge (RES).

Reverb
The blurred echoes of a room.

Root
The home note of a key or chord.

Scale
A set of notes that belong together.

Semitone
The smallest step on the keyboard: one key to the very next.

Send
How much of a track goes into a shared effect.

Sequencer
Plays stored notes in time, round and round.

Step
One box of time in the sequencer.

Sustain
The loudness a note holds at while the key stays down.

Swing
Delaying every second step a little, for a lilt.

Tie
A step that holds the previous note on instead of playing a new one.

Timbre
The tone or colour of a sound.

Track
One of the four parts, each with its own sound and steps.

Transpose
Moving notes up or down by a number of semitones.

Triad
A three-note chord built from every other note of a scale.

Trompette
The hurdy-gurdy's buzzing string (BUZZ).

Unison
One note played by several detuned voices at once.

Velocity
How hard a key is struck (from a MIDI keyboard).

Voice
One sounding note; GHOULBOX has 8 shared by the tracks.

Waveform
The shape an oscillator repeats: sine, triangle, square, saw.

Wow and flutter
The slow and fast pitch wavering of old tape (TAPE).

28 · Credits and licence

GHOULBOX © 2026 Jason Persinger. Free software under the **GPL-3.0-only** licence; the source is at `github.com/jasonpersinger/ghoulbox-fm1-dungeon-synth`.

Built entirely on **Felucca** by Leo Kuroshita / Hügelton Instruments (GPL-3.0-only). Please support the original: GitHub Sponsors (hugelton) or itch.io (hugelton.itch.io/felucca).

- Samples: Versilian Studios, VSCO-2 Community Edition and VCSL (CC0 1.0)
- Fonts: VT323 by Peter Hull and Pirata One by Rodrigo Fuenzalida and Nicolas Massi (SIL OFL 1.1)
- Felucca's engines draw on Dexed (FM6), DaisySP and Mutable Instruments' code (PHYS), klattsch (VOICE) and CrispyZebra (PHASE); see Felucca's README for the full credits
- Hardware facts in chapter 1 are from M-VAVE's FM-1 manual, in our own words.

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